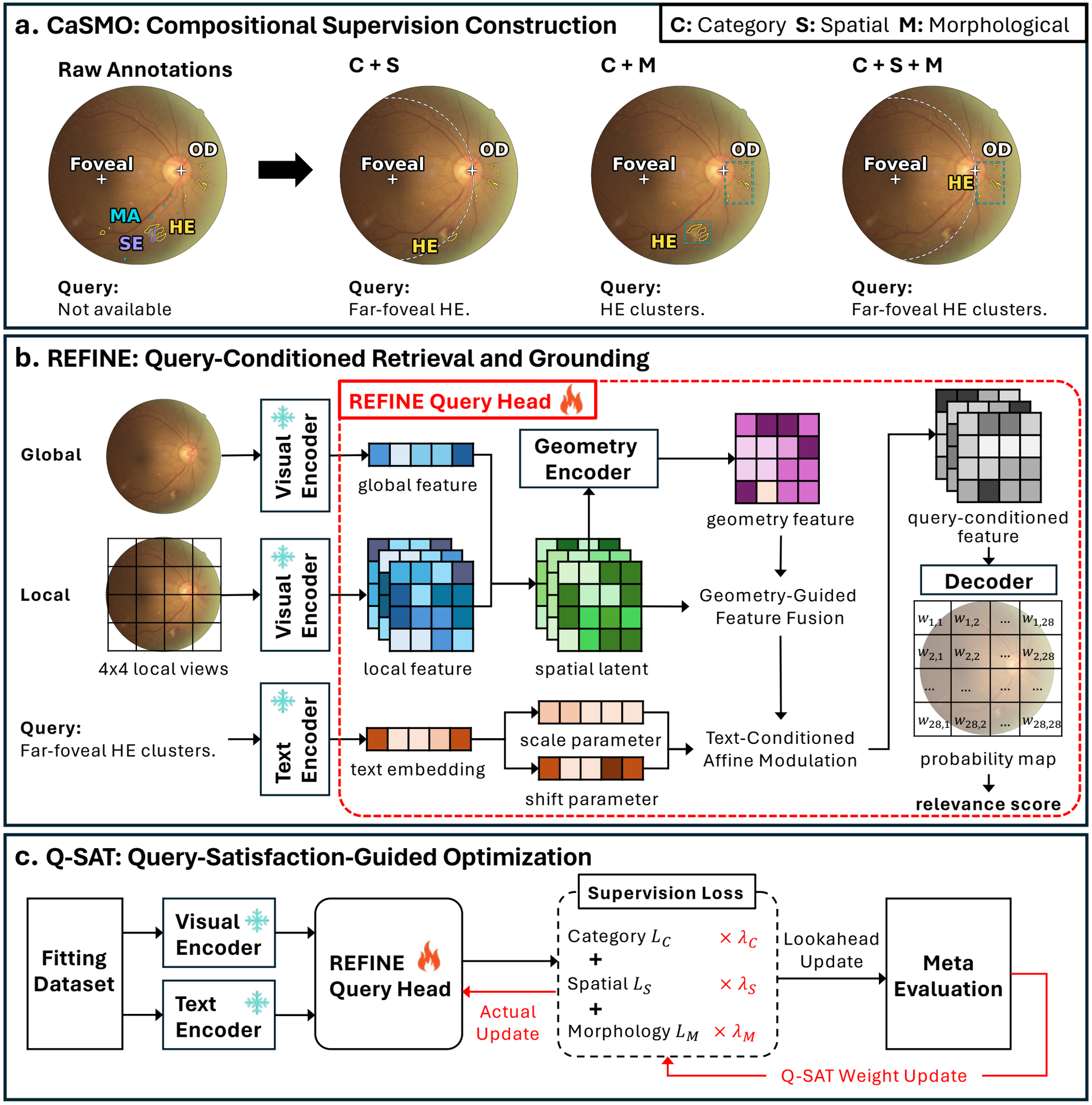

a. CaSMO: Compositional Supervision Construction
C: Category S: Spatial M: Morphological
Raw Annotations
C + S
C + M
C + S + M
Foveal
OD
MA
HE
SE
Query:
Not available
Query:
Far-foveal HE.
Query:
HE clusters.
Query:
Far-foveal HE clusters.
b. REFINE: Query-Conditioned Retrieval and Grounding
REFINE Query Head
Global
Local
Query:
Far-foveal HE clusters.
Visual Encoder
Visual Encoder
Text Encoder
4x4 local views
global feature
local feature
text embedding
Geometry Encoder
geometry feature
spatial latent
Geometry-Guided Feature Fusion
scale parameter
shift parameter
Text-Conditioned Affine Modulation
query-conditioned feature
Decoder
probability map
relevance score
c. Q-SAT: Query-Satisfaction-Guided Optimization
Fitting Dataset
Visual Encoder
Text Encoder
REFINE Query Head
Actual Update
Supervision Loss
Category L_C $\times \lambda_C$
+
Spatial L_S $\times \lambda_S$
+
Morphology L_M $\times \lambda_M$
Lookahead Update
Meta Evaluation
Q-SAT Weight Update